Email-Based Remote Server Monitoring System

Group No:6

SI NO	NAME	ROLL NO	REGISTER NO
1	ALEENA MARY V A	21	MGP24CS0021
2	ABISHEK S	6	MGP24CS006
3	ABISHEK A JOSEPH	5	MGP24CS005
4	ALVIN REJU	25	MGP24CS025

Objective

In modern computer systems, monitoring server performance is important to ensure smooth operation and prevent system failures. The Server Monitoring System with Email Alert is a project designed to monitor essential system parameters such as CPU usage and memory usage in real time.

Introduction

The system is designed to monitor system parameters such as CPU, memory, and disk usage by using a lightweight agent that collects data from multiple servers and sends it securely via HTTPS to a centralized backend built with Node.js and Express. The backend validates the data, stores it in a MySQL database, and evaluates it against predefined alert rules to detect abnormal conditions. When any threshold is exceeded, the system automatically sends email notifications to the administrator using SMTP services. Additionally, a web-based dashboard is provided to visualize system performance, monitor server status, and view alerts in real time, helping to improve system reliability and reduce downtime.

Database

SERVERS(server_id, server_name, ip_address, location, status, api_key, secret_key)

METRICS(metric_id, cpu_usage, memory_usage, disk_usage, recorded_time, server_id)

ALERTS(alert_id, server_id, alert_message, , alert_type, alert_message, severity, alert_time, resolved_time)

ADMIN(admin_id, admin_name, phone_number, email)

EMAIL LOG(email_id, alert_id, email_status, sent_time, admin_id)

Software and Tools to be Used

Programming Language: Python

Database: MySQL

Development Environment: Node.js

Operating System: Windows 11

Web Framework: Express.js

Database Connectivity: mysql2

Frontend Technologies: HTML, CSS, and JavaScript

Communication Protocol: HTTP

Email Service: SMTP / Nodemailer

Development Environment: Node.js

Diagram Designing Tool: draw.io

Design Block Diagram

Work Division Among groups

Aleena Mary V A	System monitoring (CPU, memory, disk).
Abishek S	Backend development using Node.js and database management using MySQL.
Alvin Reju	Frontend development using HTML, CSS, and JavaScript.
Abhishek A Joseph	Email alert system and documentation